

SLPC++: Teaching Software Engineering Project Courses in Industrial Application Landscapes – A Tutorial

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Abstract

Single large project courses (SLPC) with clients from industry have been established as capstone courses in many software engineering curricula. They are considered a good way of teaching industry relevant software engineering practices to students. However they are lacking in two regards. First, the results are often seen as exploratory prototypes by the client and are not intended for productive use. Second, the projects are usually not integrated into the application landscape of the client. Both of these challenges are of great practical importance in real software engineering projects. In this tutorial we cover the key factors for successfully carrying out our teaching approach called SLPC++ to address these issues. First we show how to initiate such a project course with real clients and problems. Then we describe how to carry out the project using a hybrid process model based on a combination of the Unified Process and Scrum. Finally we address the issues instructors face during such courses. Throughout the tutorial we use examples from a series of four project courses carried out in a highly successful collaboration with the Munich Airport.

1. Introduction

Project courses are an established method of teaching students realistic software engineering practices. However, these projects are primarily greenfield engineering projects. In contrast, most projects in industry are not standalone, but embedded into large industrial application landscapes, such as the communication infrastructure of a car, or the transportation system management software of an airport. In these projects, two types of problems commonly arise: (1) In the simpler case, new client systems made possible by emerging technology enablers need to be added to the application landscape, resulting at most in changes to the interface of existing systems. (2) In the more difficult case changes have to be made to the application landscape itself.

We developed a teaching approach called SLPC++ that makes it possible for students to gain industry relevant working experience by taking part in these types of projects. SLPC++ project courses are carried out in cooperation with industrial clients who want to make innovative changes to their large application landscapes. The development of the software in these courses focuses on producing code, even products, that can be used in the client's target environment.

The tutorial will give an overview of SLPC++ related problems and heuristics.

2. Definition of SLPC++

SLPC++ is organized as a one or two semester project and is designed for between 30 and 60 student participants. These students are divided into teams with sizes between five and eight members per team. All teams work on a single project topic, which is provided by the client in form of a problem statement at project kickoff. SLPC++ projects are interface and/or reengineering projects taking place inside the client's application landscape. Because on the one hand new and experimental technologies are used, but on the other hand the transition into the client's target environment also constitutes an essential project goal, a trade-off between innovation and productization must be made.

3. Specifics of SLPC++

SLPC++ courses are partitioned in the following phases: Pre-development, development, and post-development.

Pre-development phase - Preparation

SLPC++ starts with the preparation phase. At first the requirements elicitation has to be done by the instructor in cooperation with the client. Based on this elicitation a problem statement is formulated and an initial top level architecture of the system is designed.

Development phase - Project

This phase covers the main part of the project course itself. The following bullet points describe what we have found to be the best setting for the organization of this phase:

- ***Development Process***

We apply a hybrid development process during the project. In the first phase of the project we employ a version of the Unified Process adapted for the project course. In the second phase we switch to a modified agile process based on Scrum, where students ideally work full time on the project. Throughout the whole project we are using continuous integration for build and release management.

- ***Working Environment***

In SLPC++ projects, students experience two different working environments. During the first phase the students work at the university and from home. In the second phase, the whole project moves to the client's target environment. During this period the teams work full-time on the project. The focus here is on the integration of the developed software into the client's application landscape.

Post-development phase - Further transition to target environment

In contrast to traditional single large project courses the project does not end with the final presentation and handing over the demonstrated system at the client acceptance test. Instead the students are encouraged to continue participating in the transition to the final product in form of guided research, Master theses, or internships at the client.

4. Content of the Tutorial

The tutorial gives an overview over the entire organization of SLPC++ and provides instructors with the information they need to carry out such courses. The content includes topics ranging from the preparation phase of the project to the project wrap-up. We also show how all stakeholders benefit from these courses.

Overview of the tutorial content

- How to find a client & How to find a problem
- Success factors for the collaboration with the client
- Possible project types for these courses
- Detailed description of the different phases of the project
- The hybrid process model, consisting of modified versions of the Unified Process and Scrum
- Project constraints and trade-offs
- How to fit the project course into the curriculum
- Teaching strategy - what to teach and how to teach it
- Typically problems for the instructor and how to solve them
- Case studies and lessons learned

We will provide handouts for the tutorial participants which contain the information given in the tutorial as well as other material useful for the organization of SLPC++.

5. Conclusion

Every project manager in industry has these problems: new innovations, new requirements, no time. Usually we do not teach these kind of problems in university courses, especially not how to apply them to existing systems.

In this tutorial we attempt to demonstrate that teaching projects that include industrial landscapes as well as new and disruptive technologies is possible. Carrying out such project courses can be difficult. We point out some of the issues which can arise, show how to cope with them in the university setting, and present heuristics for successfully conducting SLPC++ courses. We illustrate our experiences with material from four such courses.

Our SLPC++ approach shows that it is possible to teach project courses which are deeply integrated in the application landscape of an industrial client. It offers the opportunity to create unique learning possibilities and strengthens the cooperation between university and industry.